

CRISTIANO ULONDU MENDES

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EDUCATION

PhD in Signal and Image Processing *Palaiseau, France*

Thesis: Deep learning for Synthetic Aperture Radar (SAR) data processing and analysis *2023 -- present*

Topics: speckle filtering, polarimetric SAR despeckling, tomographic SAR reconstruction,

spatially-correlated noise, unrolled algorithms, diffusion models

Supervisors: Florence Tupin, Christophe Kervazo, Loïc Denis

LTCI, Télécom Paris, Institut Polytechnique de Paris

Master's Degree in Mathematics, Data, and Learning *Paris, France*

Track: Mathematics, Modeling, Learning *2021 - 2022*

Paris Cité University

with highest honour

Bachelor's Degree in Fundamental and Applied Mathematics *Paris, France*

2020 - 2021

Paris Cité University

with high honours

SKILLS

Programming languages: Python, MATLAB, $\text{\LaTeX}$

Frameworks & Libraries: PyTorch, NumPy, SciPy, scikit-learn, OpenCV

Research areas: Deep learning, inverse problems, sparse reconstruction, variational methods, SAR imaging, polarimetry, tomography

Tools: Git, GitLab, Linux, HPC/cluster computing

Operating systems: Linux, Windows

PROJECT

Master 1 semester project: exploration of the state of the art in inverse problem-solving methods with the aim of application in medical imaging.

Demonstrated skills:

- adaptability
- technical rigor
- autonomy and involvement

EXPERIENCE

- End of studies internship *01/03/22 - 31/08/22*
Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

internship subject: Exploration of variational approaches for multi-temporal despeckling of polarimetric Synthetic Aperture Radar (SAR) images.

Tasks performed :

- bibliographic study on SAR imaging
- porting the Matlab code of a single-date SAR image denoiser (MuLoG) to Python
- proposal for an extension to polarimetric SAR images of a multi-temporal denoising scheme initially developed for single-channel SAR images (RABASAR)
- proposal of solutions to account for noise spatial correlation in the despeckling of real SAR images.

Developed skills:

- organizational skills
- autonomy
- ability to analyze scientific articles, adapt, and implement the methods described therein.

- Research engineer 15/10/22 – 31/08/23
Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

Subject: Extension of the end-of-studies internship

PUBLICATIONS

Journal Articles

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Fast tomographic SAR inversion by unrolling a greedy algorithm for sparse reconstruction,`` *submitted to IEEE Transactions on Geoscience and Remote Sensing*, June 2026. [Preprint: hal-05609504]
- C. Ulundu Mendes, E. Dalsasso, L. Denis, F. Tupin, ``POLARSAR2POLARSAR: A semi-supervised despeckling algorithm for polarimetric SAR images,`` *ISPRS Journal of Photogrammetry and Remote Sensing*, 2025. DOI: 10.1016/j.isprsjprs.2025.01.008
- C. Ulundu Mendes, L. Denis, C. Deledalle, F. Tupin, ``Robustness to spatially-correlated speckle in Plug-and-Play PolSAR despeckling,`` *IEEE Transactions on Geoscience and Remote Sensing*, 2024. DOI: 10.1109/TGRS.2024.3432180

Conferences & Workshops

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Leveraging LiDAR datasets to improve SAR tomography: a diffusion model approach,`` *14th IAPR Workshop on Pattern Recognition in Remote Sensing*, Lyon, France, Aug 2026.
- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Reconstruction 3D en tomographie radar : apprentissage profond basé sur un Matching Pursuit déroulé,`` *GRETSI 2025*, Strasbourg, France, Aug 2025.
- C. Ulundu Mendes, L. Denis, F. Tupin, ``Schéma Plug-and-Play robuste à la corrélation spatiale du speckle pour le filtrage des données SAR polarimétriques,`` *GRETSI 2023*, Grenoble, France, Aug 2023.